

Task-13

15/10/25

Finding the winning strategy in a card game

AIM: To implement a python program that simulates a card game b/w two players and determine the winning strategy using of drawn cards

ALGORITHM:

1. Start the program
2. Create a deck of cards
3. Shuffle the deck randomly
4. Each player draws a fixed number of cards.
5. Define the strategy
6. Compare the chosen cards of both players.
 - If player 1's card > player 2's card $\rightarrow$ player 1 scores
 - Else if player 2's card > player 1's card $\rightarrow$ player 2 scores
7. Repeat for all rounds
8. The player with the highest score wins the game
9. Display input, output and a final result
10. End the program

program:

```
import random
```

```
suits = ["Hearts", "Diamonds", "Clubs", "Spades"]
```

```
values = list(range(1, 14))
```

```
deck = [value, suit for suit in suits for value in values]
```

```
random.shuffle(deck)
```

```
player1-hand = deck[:5]
```

```
player2-hand = deck[5:10]
```

```
print("Player 1 hand:", player1-hand)
```

```
print("Player 2 hand:", player2-hand)
```

```
def play_highest_card(hand, card):
```

```
    highest = max(hand, card)
```

```
    hand.remove(highest)
```

```
    return highest
```

```
p1_score, p2_score = 0.0
```

```
print("Welcome to the game of cards")
```

```
for i in range(5):
```

```
    p1_card = play_highest_card(player1-hand, card)
```

```
    p2_card = play_highest_card(player2-hand, card)
```

```
    print(f"Round {i+1} Player 1: {p1_card}")
```

```
    print(f"Player 2: {p2_card}")
```

```
    if p1_card > p2_card:
```

```
        print("Winner: Player 1")
```


Sample Output:

Player 1 Hand: (13, 'Hearts'), (2, 'Clubs'), (10, 'Spades'),
(5, 'Diamonds'), (7, 'Clubs')

Player 2 Hand: (9, 'Hearts'), (12, 'Diamonds'), (3, 'Clubs'),
(11, 'Spades'), (6, 'Hearts')

Sample Output:

Round 1 - Player 1 -> (13, 'Hearts'), Player 2 -> (12, 'Diamonds')
Winner: Player 1

Round 2 - Player 2 -> (10, 'Spades'), Player 2 -> (11, 'Spades')
Winner: Player 2

Round 3 - Player 3 -> (7, 'Clubs'), Player 2 -> (9, 'Hearts')
Winner: Player 2

Round 4 - Player 1 -> (5, 'Diamonds'), Player 2 -> (6, 'Hearts')
Winner: Player 2

Round 5 - Player 1 -> (2, 'Clubs'), Player 2 -> (3, 'Clubs')
Winner: Player 2

Q/P ✓ 0.5/10/15


```

    if p2 - card(0) > p1 - card(0):
        print("winner: played 2")
        p2 - score + 1

```

```

    else:
        print("result: draw")

```

```

    print("with --- finaly Result ---")
    print("player 1 score: " + p1 - score)
    print("player 2 score: " + p2 - score)

```

```

    if p1 - score > p2 - score:

```

```

        print("player 1 win the game with winning strategy")

```

```

    elif p2 - score > p1 - score:

```

```

        print("player 2 win the game with winning strategy")

```

```

    else:

```

```

        print("the game is a draw!")

```

WILL TECH CSE	
FEED	13
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	15
TOTAL (20)	
SIGN WITH DATE	

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15/05/2025

RESULT:

Thus, the finding winning strategy in a card game is executed successfully

R. Jany
15/05/2025